BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD CAMPUS

EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2025-26

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Semi-Controlled converters and Three Phase Controlled Converters, Buck converter

Q1. A single-phase half controlled symmetrical bridge converter is fed from 240V, 50Hz supply. Neglecting voltage drop across the devices, determine average and RMS voltage at firing angles 0^0, 90^0, and 180^0. If the RL load is highly inductive, with an R=10Ω. Determine the required device ratings.

Q2. A centre-tapped fully controlled converter is supplied at 120V line to neutral. Determine the mean load voltage for firing angles of 0^0, 30^0, 60^0, and 90^0, assuming an RL load with load current to be continuous and ripple free with a constant 1.5V drop on each thyristor in the forward biased condition.

Q3. A three phase fully controlled bridge converter is used for charging a battery with an EMF of 110V and an internal resistance of 0.2Ω. A large inductor is connected in series with the load to maintain constant charging current of 10A,

(i) (a) Assuming ideal condition to all thyristors, compute the firing angle for AC line voltage of 220V (b) If each thyristor has a voltage drop of 2V across them, compute the firing angle.

(ii) (a) For the purpose of delivering power of 1.1kW from the battery to the 3-phase system the converter firing angle is increased to 150^0. Assuming ideal condition of thyristors and for the same value of DC average current of 10A, compute the peak AC line voltage (ideal condition).

Q4. In a buck converter R_{load} = 5Ω, L = 10mH, f = 1kHz, Vs = 100V, D = 0.5, then determine (i) DC component of the load current (ii) Peak to peak ripple current (iii) If the frequency is increased to 4kHz how will the ripple current magnitude be affected (iv) If rather than changing frequency the load inductance is increased to 40mH then what percentage will be ripple current?

Q5. Design a buck converter for telecommunication power supplies which requires high current and low voltage. For a 3.3V of input supply an output voltage of 1.2 V is obtained. For an input average current of 2.2A, switching frequency of 500KHz and a peak to peak variations for the inductor current not exceeding 40 % of the average inductor current, determine the value of inductor and capacitor filter used. Consider a safety factor of 1.6 and output voltage ripple factor of 2%.